

Department of Computer Science

Examination paper for IT3708 - Bio-inspired Artificial Intelligence

Examination date: May 15, 2024

Examination time: 15:00-16:30 h

Permitted examination support material: D / No printed or hand-written support material is allowed. A specific basic calculator is allowed.

Academic contact during examination: Xavier F. C. Sánchez Díaz

Phone: 93 07 01 18

Academic contact present at the exam location: NO

OTHER INFORMATION

Get an overview of the question set before you start answering the questions.

Read the questions carefully and make your own assumptions. If a question is unclear/vague, make your own assumptions and specify them in your answer. The academic person is only contacted in case of errors or insufficiencies in the question set. Address an invigilator if you suspect errors or insufficiencies. Write down the question in advance.

No hand drawings: This exam does not include hand drawings. If you receive hand drawing sheets, this is by mistake. **You will not be able to submit the sheets, and they will not be graded.**

Weighting: Each “multiple choice” question has exactly one answer and is worth 1 point, for a total of 18 points. The “written answer” questions are worth a total of 12 points. For each “written answer” question, the maximum score is given with the question. The total number of points for the exam is 30.

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspira. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top righthand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers in the archive in Inspira. Be aware that it may take a working day until any hand-written material is available in the archive.

- 1 Complexity theory includes theory about NP-complete and NP-hard problems. Which of the following statements is **false**?

Select one alternative:

- ☐ A problem is NP-complete if it is at least as hard as any problem in the class NP-hard. ✓
- ☐ It is not known whether the complexity classes P and NP are the same or not.
- ☐ If a problem is NP-hard, it often makes sense to design and implement a bio-inspired algorithm for it, to obtain a heuristic solution to the problem.
- ☐ A problem can be both NP-complete and NP-hard.
- ☐ For an NP-complete problem, some problem instances may be solvable in polynomial time.

Maximum marks: 1

- 2 Mutation is, in addition to recombination, the main variation operator in evolutionary algorithms (EAs). Evaluate the following statements about mutation and pick the one that is **false**.

Select one alternative:

- ☐ Evolution strategies historically used mutation only, there was no crossover.
- ☐ A mutation-only evolutionary algorithm can work well – crossover is not needed.
- ☐ The goal of the mutation operator is to merge information from two or more parents. ✓
- ☐ Mutation has a theoretical as well as practical role: it ensures that an EA can reach any state in the search space from any other state.
- ☐ Mutation can, for a given individual representing a search space state, move it to any other arbitrary state in the search space.

Maximum marks: 1

- 3 Recombination (or crossover) is one of the variation operators in evolutionary algorithms (EAs). Evaluate the following statements about crossover and pick the one that is **false**.

Select one alternative:

- ☐ The goal of the crossover operator is to merge information from two or more parents
- ☐ A crossover-only genetic algorithm works well in reaching an optimum – mutation is not needed. ✓
- ☐ Recombination is, in addition to mutation, the main variation operator in genetic algorithms.
- ☐ The crossover operator is typically stochastic.
- ☐ Crossover traditionally plays a greater role in genetic algorithms than in evolution strategies.

Maximum marks: 1

- 4 Selection, replacement and elitism play key roles in evolutionary algorithms. Among the following statements, pick the one that is **false**.

Select one alternative:

- ☐ A potential disadvantage of using elitism is that it may lead to premature convergence.
- ☐ Tournament selection computes the fitness over the whole population in each tournament. ✓
- ☐ Rank-based selection is less computationally efficient than tournament selection.
- ☐ Tournament selection is more computationally efficient than both fitness proportional selection and rank-based selection.
- ☐ An advantage of tournament selection over many other forms of selection is its computational efficiency.

Maximum marks: 1

- 5 Evaluate the following statements about the population management in an evolutionary algorithm and pick the one that is **true**.

Select one alternative:

- ☐ How selection is performed strongly depends on which crossover method is used.
- ☐ Selection in evolutionary algorithms is often performed by means of local search.
- ☐ Selection from the population does not depend on the chromosome representation. ✓
- ☐ How selection and replacement work together does not need to be considered.
- ☐ When you perform selection in a genetic algorithm, there is no need to do replacement.

Maximum marks: 1

6 Evaluate the following statements about the island model in bio-inspired AI, and pick the one that is **false**.

Select one alternative:

- ☐ Each island can use a different evolutionary algorithm from the other islands.
- ☐ A major benefit of the island model is how it naturally can utilize parallel computing.
- ☐ Different hyperparameter settings can be used for different islands.
- ☐ In the island model, the most typical evolutionary approach used is memetic algorithms. ✓
- ☐ Each island evolves semi-independently from the other islands – there is some exchange of individuals between islands.

Maximum marks: 1

7 Evaluate the following statements about real-valued genetic algorithms and pick the one that is **true**.

Select one alternative:

- ☐ In order to create a real-valued GA from a binary (or pseudo-boolean) GA, both initialization and crossover need to be changed.
- ☐ In order to create a real-valued GA from a binary (or pseudo-boolean) GA, both crossover and mutation need to be changed.
- ☐ In order to create a real-valued GA from a binary (or pseudo-boolean) GA, only crossover needs to be changed.

- ☐ In order to create a real-valued GA from a binary (or pseudo-boolean) GA, both initialization and mutation need to be changed. ✓
- ☐ In order to create a real-valued GA from a binary (or pseudo-boolean) GA, only selection needs to be changed.

Maximum marks: 1

- 8 Evaluate the following statements about differential evolution and genetic algorithms (GAs) and pick the one that is **true**.

Select one alternative:

- ☐ Among other bio-inspired AI methods, differential evolution is most similar to ant colony optimization.
- ☐ Differential evolution can be used with different population sizes. ✓
- ☐ Differential evolution is primarily for combinatorial problems, while GAs work well for both real-valued and combinatorial problems.
- ☐ The main difference between a real-valued GA and differential evolution is that only the latter typically uses mutation.
- ☐ Differential evolution takes its name from difference equations and differential equations.

Maximum marks: 1

- 9 Evaluate the following statements about niching in genetic algorithms (GAs). Which option among the following is **not** a good rationale for niching? (which option is **false**)

Select one alternative:

- ☐ Niching including crowding algorithms are useful for optimizing multimodal functions.
- ☐ Niching including crowding are inspired by species and formation of species in biology.
- ☐ Niching methods typically increase the chance of premature convergence compared to the simple GA (everything else being equal). ✓
- ☐ In deterministic crowding , children always replace their parents in the population while in the simple GA this is not necessarily the case.

- ☐ Crowding algorithms will typically maintain entropy at a higher level than a simple GA (everything else being equal).

Maximum marks: 1

- 10** Evaluate the following statements about Markov chain theory as applied to genetic algorithms (GAs) and pick the one that is **false**.

Select one alternative:

- ☒ Homogenous Markov chains are needed to exactly model adaptive GAs, where state transition probabilities change over time. ✓
- ☐ When we make the Markov assumption for a GA, the GA's next state only depends on its current state.
- ☐ If a GA can be formalized by an irreducible and periodic Markov chain, that Markov chain's stationary and occupancy distributions are equal.
- ☐ The transition matrix plays an essential role when using Markov chains to analyze GAs.
- ☐ While continuous-time Markov chains exist, discrete-time Markov chains are most suitable for modeling most GAs.

Maximum marks: 1

- 11** Constraints often appear in optimization problems from industry. Which of the following statements about constrained optimization using evolutionary algorithms (EAs) is **false**?

Select one alternative:

- ☒ Constraint satisfaction problems cannot be solved by EAs as they have no fitness function, which is a crucial element for any solver in evolutionary computation. ✓

Constraint handling techniques in evolutionary computation include using a penalty ☐ function (linear, quadratic or exponential) as well as boolean functions that use distance to the feasible region to alter the overall fitness of a solution.

- ☐ The Traveling Salesperson Problem, the Eight Queens and many scheduling problems are examples of problems with constraints that can be solved using EAs.

A constraint optimization problem can be solved using an evolutionary algorithm if there is ☐ some sort of constraint handling in the algorithm. Counting the number of constraint violations and using it as a fitness penalty is an example of constraint handling.

- ☐ A free optimization problem can be solved by an EA as long as there is a fitness function, for example as in the Ackley or Rosenborg functions.

Maximum marks: 1

- 12** In the Ant Colony Optimization (ACO) metaheuristic, each individual simulates the path an ant takes by walking randomly through the search space. In the end, valid paths to the solution are returned. Which of the following statements regarding ACO is **false**?

Select one alternative:

- ☐ ACO is an iterative procedure in which the interaction of multiple ants makes finding the solution possible.
- ☐ ACO was designed for combinatorial problems that are based on graphs, but can be adapted to the continuous domain by discretizing the search space into "bins".

An ant generates a complete tour of nodes using a transition rule that is probabilistic. The ☐ transition prioritizes high pheromone trails (smelly paths) and shorter distances to find an optimal path.

The pheromone update and evaporation parameters are of utter importance when it comes ☐ to the exploration and convergence. Their values need to be carefully set up as they depend on each other.

☐ To ensure convergence to the optimum, the pheromone update parameter needs to be set to a smaller value than the pheromone evaporation parameter. ✓

Maximum marks: 1

- 13** As in solvers in machine learning, many of the metaheuristics and algorithms in evolutionary computation need some sort of parameter setting. Which of the following statements regarding parameters is **false**?

Select one alternative:

- ☐ Encoding mutation rates as part of an individual's chromosome is a form of self-adaptive parameter control.
- ☐ Parameters can be set up and adjusted before the run and during the run, which we call parameter tuning and parameter control, respectively.
- ☐ Parameter tuning is performed by the user, and it relies on the user's knowledge about the metaheuristic and about the problem being solved.
- ☐ Setting a rule for increasing the mutation rate every k generations is a form of deterministic parameter control.
- ☐ Increasing the mutation rate after every k successful mutations in Evolutionary Strategies is a form of self-adaptive parameter control. 

Maximum marks: 1

- 14 Search and convergence in an evolutionary algorithm might be dependent on the parameters you use. Which of the following statements about parameters is **true**?

Select one alternative:

Generalized Crowding is a technique to scale the importance of crowding distance ☐ depending on the number of neighbors a given solution has. This is the basis for NSGA-II, in which the scaling parameter is crucial for the non-dominated sorting.

The 1/5 rule in Evolutionary Strategies is a form of adaptive parameter control, since it ☒ resets the standard deviation (or noise) of mutation rates after every k iterations depending on the success rate of evolution.

Multi-objective optimization is inherently more difficult than single-objective optimization, as ☐ the objectives may be conflicting. To effectively handle this conflict, we use parameter control in NSGA-II by letting the algorithm select the number of objectives that best fits the problem.

Non-deterministic parameter control refers to online modification of an EA's parameters, ☐ and it depends entirely on the algorithms behavior as the user has no control on the search. This is usually achieved by incorporating the desired parameters to the individual so that they are "evolved" along the rest of the chromosome.

In a complicated fitness landscape, it is desired to vary the parameters through the run. ☐ This is done by running the algorithm multiple times and observing the performance, and then providing a list of values which will be used during runtime. This is known as parameter control.

Maximum marks: 1

- 15 The real world contains scary situations in which multiple, conflicting objectives need to be optimized at the same time. Which of the following statements about multi-objective optimization is **false**?

Select one alternative:

☐ When comparing two solutions, we cannot assume that one of them is better than the other just by looking at a single objective in isolation.

☐ By looking at the Pareto set in the objective space, one can obtain the Pareto-optimal front which is used to select the final solution.

☐ A solution x is said to dominate another solution y if x is not worse than y in all objectives.

The ☒ individual's genotype and phenotype are part of the objective space. ✓

☐ The set of all non-dominated solutions is known as the Pareto-optimal set.

Maximum marks: 1

- 16 One of the most successful algorithms in multi-objective optimization is NSGA-II. Which of the following statements about NSGA-II is **false**?

Select one alternative:

- ☐ NSGA-II uses a fixed population size N . When selecting survivors, the N best individuals are chosen from the $2N$ available, sorting only those solutions in the last rank.
- ☐ NSGA-II uses a ranking system in which each solution gets a rank that is based on the number of solutions it is dominated by.

To sort individuals, NSGA-II uses crowding distance. This distance (which is more like a crowd-index) assigns a better score to those solutions in a lonelier neighborhood, as opposed to the solutions in highly crowded areas of the objective space.

- ☐ NSGA-II is inherently multimodal, as it uses niching to preserve diversity—each rank contains several niches obtained by using crowding distance.

NSGA-II creates offspring using the crowding tournament selection operator. Using the crowding distance (calculated beforehand for all individuals in the population) allows the NSGA-II to do a fast non-dominated sorting. ✓

Maximum marks: 1

- 17 In an EA, the choice of representation of an individual is problem dependent. Which of the following algorithms, in their traditional version, is **not** tailored for continuous optimization?

Select one alternative:

- ☐ Evolutionary Strategies (ES)
- ☐ The Continuous Genetic Algorithm (CGA)
- ☐ Pseudo-Boolean Stochastic Local Search (SLS) ✓
- ☐ Particle Swarm Optimization (PSO)
- ☐ Differential Evolution (DE)

Maximum marks: 1

- 18 Multimodal optimization deals with optimization problems where multiple optimal solutions exist. To achieve this, an EA needs some sort of diversity preservation. Which of the following techniques is **worst** at preserving diversity?

Select one alternative:

- ☐ Fitness Sharing
- ☒ Elitism
- ☐ Tournament selection and Stochastic Universal Sampling
- ☐ Migration in island models
- ☐ Crowding and Nicheing



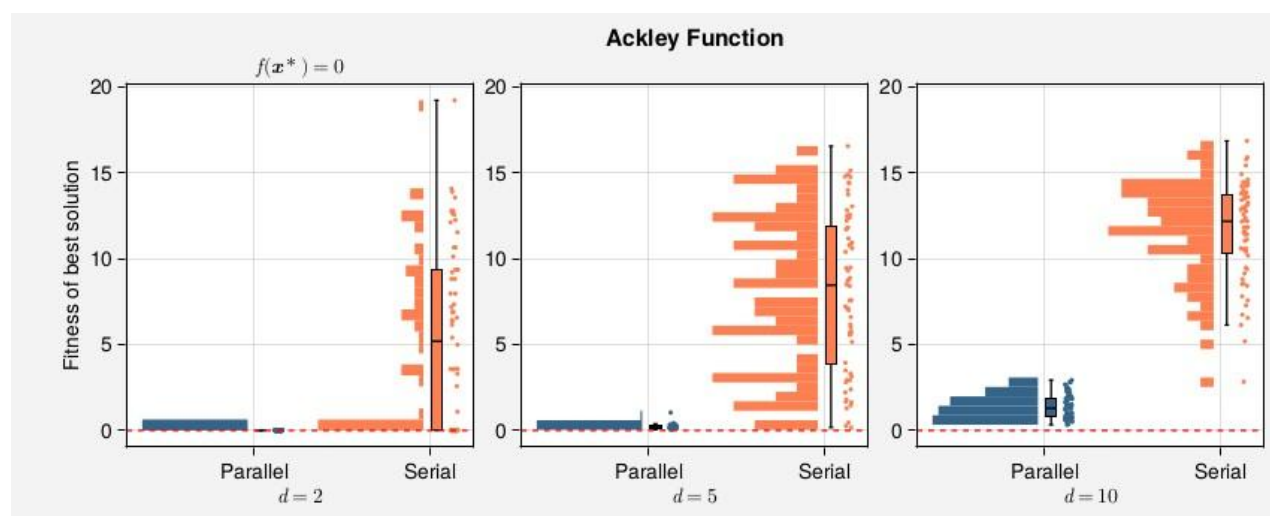
Maximum marks: 1

- 19 In this question we study the performance of a parallel island generational GA versus a traditional generational GA run repeatedly (serially).

The GA considered has a population size of 30 individuals, with Gaussian mutation with standard deviation of 0.1, and uniform crossover. The population was initialized uniformly at random, and we used rank-based parent selection.

The parallel experiment (in blue) was run using 64 islands in an island model of a GA, while the serial experiment is the same GA (with the same parameters) but restarted 64 times.

Consider the following group of plots.



In these plots, we compare the two experiments: Parallel (blue) versus Serial (orange), for three different versions of the Ackley function; each corresponding to a different number of dimensions,

d. The y-axis in each plot shows the fitness of each experiment (a dot), and a histogram and box plot help quantify the spread of the fitness in their respective results. The optimal fitness in all cases is 0, which is highlighted with a red, dashed line.

1. What can be inferred from the two experiments?
2. How do their performances compare in terms of convergence?
3. Why does the serial approach seem to get results with high variation?

Write **two-three sentences for each question** (6p total, 2p each).

Fill in your answer here

Format ▾ | **B** *I* U x_2 x^2 I_x | | | |

| | Σ |

NB! This is a very open question, we expect a brief description of what is happening and what is being compared. We have included many aspects of what could be mentioned in the answers.

One of the approaches is better than the other. The parallel island model gets better results and more consistently. The serial approach fails to converge on multiple occasions. The communication in the parallel island model helps the convergence. The problem seems to become harder with a higher number of dimensions, **d**.

How do their performances compare in terms of convergence?

The serial approach (orange) has much more variation. The parallel approach (blue) converges to the solution more consistently. The parallel approach gets better results, that is, closer to the optimal. The serial approach is inconsistent and its distribution has a higher spread.

Why does the serial approach seem to get results with high variation?

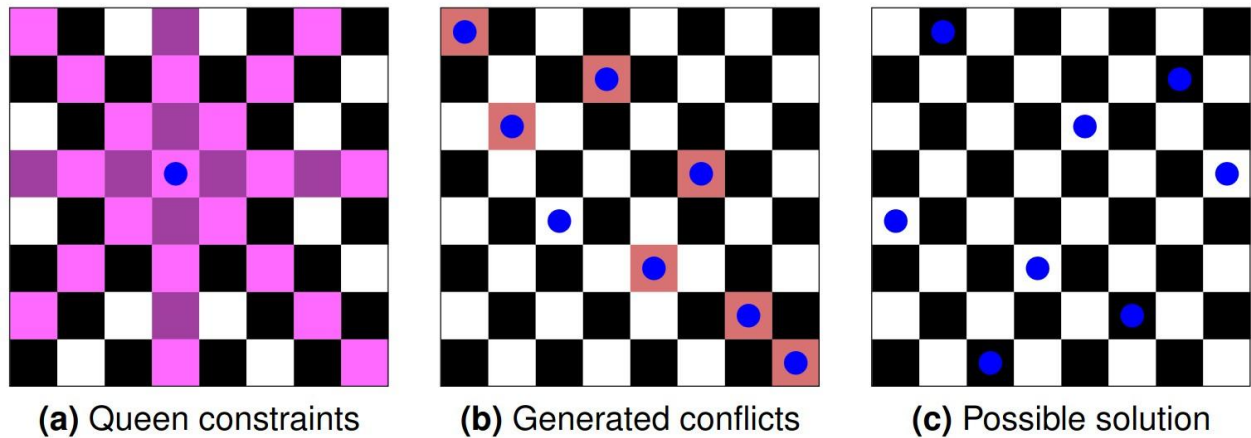
The high variation is due to the restart. We do not know the number of generations, but it seems it is not enough for the GA to converge. This is probably because the searchspace is difficult to traverse. In contrast, the parallel approach has a much more consistent output since 64 populations are exchanging individuals. The exchanged individuals help to guide the search in their new population.

Words: 0

Maximum marks: 6

- 20** We consider the 8-queens optimization problem, where 8 queens need to be placed on a chess board in an arrangement so that no queen checks any other.

Consider the following figure.



Subfigure (a) shows the constraints (in pink) imposed by the placement of a single queen piece (in blue). (b) highlights the conflicts arising from a possible configuration of the board, and (c) illustrates one possible solution with no conflicts.

Assume you will solve the 8-queens problem using an evolutionary algorithm.

1. What is a good representation for an individual?
2. What would you use as your fitness function?
3. What kind of mutation would be suitable in this case? Give 2 examples.

Write **up to four sentences for each question** (6p total, 2p each).

Fill in your answer here

Format **B** *I* U x_2 x^2 $\frac{I_x}{x}$

Σ

Words: 0

Maximum marks:
6